

FYP Pre-Start Prep — Progress Tracker

LLM-Based Interface for Robotics Systems using UR5e | Week 4 onwards | simulation moved to MuJoCo
Status snapshot as of Sunday 12 July 2026 (mid Week 4) · Supervisor: Prof Cheah · Prep period 16 Jun – 4 Aug 2026

DONE

Finished and verified

IN PROG

Started / partially built

TO DO

Not started yet

Where you stand. The foundation (Weeks 1–3) and the Architecture-B *data-capture* half are done and slightly ahead of schedule. The only item still open *this* week is the Open X-Embodiment reading. Not yet started: **all of Architecture-B training** (Week 5) and **all of the Architecture-A modular pipeline** (Week 6) — the two halves your whole comparison rests on. One code-level gap to close early: **src/fyp/transforms.py** is empty.

WEEK 4 · 7–13 Jul 2026

Deep VLA reading + MuJoCo simulation

21 done 1 in progress 2 to do · 24 items

PLAN NOTE Simulation moved from ManiSkill3 to **MuJoCo**. You built the sim stack directly in MuJoCo (mujoco_controller, ik, sim_server/client, gripper attach) — lighter, fully self-owned, and it gives you an RGB-D camera + physics that URSim lacks, which the Week-6 perception work depends on.

Mon 7 Jul — RT-2 review + transition to OpenVLA

DONE

Re-read RT-2 notes: what the action space is exactly

DONE

Map RT-2's action space onto what your URController accepts

DONE

Read OpenVLA introduction and related work

RT-1 and RT-2 already deep-read with architectural walkthroughs in Week 3.

Tue 8 Jul — OpenVLA architecture

DONE

OpenVLA Sec 3–4: Llama-2 backbone, DINOv2 + SigLIP vision encoders

DONE

Note action representation and chunking; compare architecturally to RT-2

Wed 9 Jul — OpenVLA training and eval

DONE

OpenVLA Sec 5–6: Open X-Embodiment training subset, LoRA fine-tune recipe, limitations

You confirmed OpenVLA is read. The LoRA recipe is the one you'll mirror with SmoVLA.

Thu 10 Jul — Open X-Embodiment

TO DO

Read the OXE paper / data card (Padalkar et al. 2023); action/observation format conventions; check UR5 inclusion

The one Week-4 reading item still outstanding.

Fri 11 Jul — MuJoCo simulation setup (was ManiSkill3)

DONE

Install MuJoCo; load the UR5e model (Menagerie ur5e.xml + scene.xml)

DONE

Launch the viewer; verify rendering (X11 for viewer, EGL for headless)

DONE

Attach the Robotiq 2F-85 via mjSpec → scene_gripper.xml

Evidence: ur5e_setup_mujoco.png, assets/mujoco/. ManiSkill3 was trialled (maniskill3_setup.png) then dropped.

Sat 12 Jul — Fixed-camera view (today)

- DONE** UR5e scene with a fixed third-person camera (matches the real eye-to-hand rig)
- DONE** Offscreen mujoco.Renderer captures RGB frames (240×320)
- IN PROG** RGB-D / depth output
Depth render touched (test_depth.png) but the depth-to-3D conversion itself is Week 6 work.
- DONE** Save a few frames as PNG to verify

Sun 13 Jul — Demonstration recorder

- DONE** DemoRecorder captures joints, TCP pose, gripper state, image at ~20 Hz (RateControl)
- DONE** save_episode() writes per-field arrays to HDF5
- DONE** Replay path: replay_episode.py reads HDF5 back to MP4
- DONE** Test by scripting a pick motion and recording it (record_replay_episode.py → ep_001)
- DONE** Visualise the trajectory in matplotlib; verify images (tcp_trajectory.png, ep_001_check.png)
- DONE** Bonus, beyond the plan: teleop-and-record host/client over TCP sockets (sim_server + sim_client), thread-safe command queue, a damped-least-squares Jacobian IK solver (ik.py), and the gripper-composition script (attach_gripper.py)

CODE GAP `src/fyp/transforms.py` is empty. The pose helpers (`pose_to_T`, `T_to_pose`, `pose_trans`) aren't in the tree at all — they lived in an old `square.py` that was deleted from git (recoverable from history, commit 99541bf). Separately, the quat-to-rotvec conversion is duplicated across `ik.py`, `mujoco_controller`, and `sim_server`. Recover the first set and centralise the second into `transforms.py` before Week-6 hand-eye calibration reuses it.

WEEK DELIVERABLES

- DONE** OpenVLA architecture understood (can explain it to a friend)
- DONE** MuJoCo working with a UR5e and a fixed camera (replaces 'ManiSkill3 working')
- DONE** DemoRecorder working; at least one episode saved and replayed
- TO DO** Open X-Embodiment paper read (only remaining Week-4 item)

WEEK 5 · 14–20 Jul 2026

SmoVLA + LeRobot (Architecture B training)

0 done 0 in progress 12 to do · 12 items

PLAN NOTE Data capture (DemoRecorder) is already done from Week 4 — this week is the **training half** of Architecture B. Add one bridging piece that isn't in the original plan but your architecture doc lists as 'planned': an **HDF5 → LeRobot/RLDS converter** that turns your recorded episodes (absolute states, axis-angle) into training format (action deltas, Euler). Nothing consumes your episodes yet — this is the missing link between Week-4 data and Week-5 training.

Mon 14 Jul — LeRobot installation

- TO DO** Clone huggingface/lerobot; install smolvla extras; log into HF Hub; explore the directory structure

Tue 15 Jul — SmoVLA paper (first half)

- TO DO** Sec 1–3: ~450M params (vs 7B OpenVLA), SmoVLM-2 backbone + Action Expert, async inference stack

Wed 16 Jul — SmolVLA paper (second half)

TO DO Sec 4–6: training data source, key ablations; write a 1-page summary in your own words

Thu 17 Jul — LeRobot SmolVLA tutorial

TO DO Open the fine-tuning tutorial; download the dataset; print shape + episode count; visualise a few frames

Fri 18 Jul — Cloud GPU training setup

TO DO Modal Labs or Colab Pro; training script on the cloud GPU; confirm dataset access; 1-step dry run
8GB local fine-tuning is optional; cloud is the planned path (don't fight OOM).

Sat 19 Jul — Run the fine-tune

TO DO Launch the full run (1–3 h); monitor loss curves; save the checkpoint; note cost + time

Sun 20 Jul — Inference + RTC paper

TO DO Load the checkpoint; run inference on a held-out episode; predicted vs ground-truth

TO DO Read every line of the training script; read RTC (Black et al. 2025): async inference + action chunking

WEEK DELIVERABLES

TO DO Fine-tuned SmolVLA checkpoint saved (cloud GPU)

TO DO SmolVLA paper understood + 1-page summary

TO DO RTC paper read

TO DO Inference working: predicted vs ground-truth comparison

WEEK 6 · 21–27 Jul 2026

Perception + modular pipeline + LLM planner (Architecture A)

0 done 0 in progress 15 to do · 15 items

PLAN NOTE Prototype the whole modular slice in **MuJoCo**, not **URSim**. MuJoCo gives you the RGB-D camera, depth, and gripper physics this stack needs; URSim (Polyscope) has none of that. Build primitives on the shared controller interface (URControllerMuJoCo now, URController on real hardware later) so nothing needs rewriting for deployment. This is Architecture A in full — the pipeline the project compares against the VLA, so it is now central, not peripheral.

Mon 21 Jul — Open-vocabulary detection

TO DO Run OWL-ViT / GroundingDINO on a sample image; free-form text query → boxes + confidence; threshold; visualise

The crux vs closed-set YOLO: an open-vocab detector can name and localise novel objects.

Tue 22 Jul — Depth to 3D position

TO DO Box-centre pixel + depth (RGB-D) + camera intrinsics → camera-frame XYZ; small NumPy function

You already render depth in MuJoCo — feed that straight in here.

Wed 23 Jul — Hand-eye calibration

TO DO Fixed-camera (eye-to-hand) transform; object-in-camera → object-in-base (one 4×4 multiply); NumPy
Reuses transforms.py (currently empty — populate it first). Same idea as TF on your Tank project.

Thu 24 Jul — Scripted motion primitives

TO DO pick(xyz): above → descend → close → lift. place(xyz): above → descend → open → retract

TO DO Build on the shared controller interface; test pick-and-place in MuJoCo with a hardcoded object position

Fri 25 Jul — Anthropic SDK + tool calling

TO DO API key; hello-world Claude call; read tool-use docs; define pick/place/move as tools; note rate limits + cost

Sat 26 Jul — LLM planner over primitives

TO DO System prompt describing skills + detected objects; LLM returns an ordered tool-call plan (structured)

TO DO Parse the plan and dispatch to primitives; test 'put the red block in the bin' → detect → plan → execute

Sun 27 Jul — Closed-loop modular slice + write-up

TO DO Wire the full slice: capture frame → open-vocab detect → depth-to-3D → object positions → LLM plan → execute in MuJoCo

TO DO Add a re-detect step before each grasp (closed loop); document detection-stage vs grasp-stage failures; update README

Open-loop first; add the re-detect step only once it runs end-to-end.

WEEK DELIVERABLES

TO DO Open-vocabulary detector returning boxes on test images

TO DO Depth-to-3D + hand-eye transform implemented (NumPy)

TO DO Scripted pick/place primitives working in MuJoCo

TO DO LLM planner sequencing primitives via tool calls

TO DO Closed-loop modular slice running end-to-end in simulation

WEEK 6.5 · 28 Jul – 4 Aug 2026

Refinement, consolidation, supervisor sign-off

0 done 1 in progress 11 to do · 12 items

Mon 28 – Tue 29 Jul — Refine the proposal

TO DO Re-read the drafted proposal end to end; check against NTU EEE required format

TO DO Update the architecture diagram to match what you built (MuJoCo); tighten the methodology with concrete prep details

Wed 30 Jul — Send for sign-off

TO DO Send the proposal to the supervisor; prepare 3–5 questions; confirm VLA choice, lab GPU access, budget

Thu 31 Jul — Consolidate the codebase

TO DO Tidy repo + README; ensure URController, DemoRecorder, detector, planner each have a runnable demo; write a 'how to run each piece' note

Fri 1 – Sat 2 Aug — Catch up + revise

TO DO Absorb any slippage from Weeks 1–6; revise on early supervisor feedback; re-read the 5 core papers

Sun 3 – Mon 4 Aug — Final prep for FYP start

TO DO

Write a 1-page 'current state' summary for FYP Week 1; draft a priority list for the first two FYP weeks; set a weekly review cadence

WEEK DELIVERABLES

TO DO

Proposal refined and sent for sign-off

TO DO

Codebase consolidated: URController, DemoRecorder, detector, planner all runnable

TO DO

Fine-tuned SmolVLA checkpoint (tutorial dataset)

TO DO

Closed-loop modular slice working in simulation

IN PROG

Notes on at least 5 deeply-read papers (RT-1, RT-2, OpenVLA done; OXE, SmolVLA, RTC remaining)

TO DO

1-page current-state summary + Week 1 priority list